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Ridge Regression and Multicollinearity

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Abstract

This project studies ordinary least squares (OLS), numerical conditioning, multicollinearity, and ridge regularisation using the UCI Real Estate Valuation dataset. The OLS estimator is computed through the normal equations, QR decomposition, and singular value decomposition (SVD). The numerical results are compared through fitted-value metrics and matrix diagnostics. The modelling design contains 414 observations and, after the implemented categorical expansion and intercept treatment, 17 columns. The revised full-rank design matrix has a reported condition number of approximately 2.3840×10^7, while the corresponding Gram matrix has a condition number of approximately 5.6835×10^{14}. The three OLS solution approaches produce practically identical predictive metrics, illustrating agreement of the computed least-squares solutions while also motivating care when solving ill-conditioned normal equations. Ridge regression is then examined through coefficient paths and conditioning diagnostics. A five-fold cross-validation procedure with random state 42 selects $\lambda \approx 35.2971$ from the evaluated grid. Its mean cross-validated MSE is 86.4162, compared with 87.0110 for OLS under the saved fold-comparison experiment. The improvement is modest; the report therefore interprets regularisation as a stabilising trade-off rather than claiming a dramatic predictive gain. All numerical values and plots in the results chapter are drawn from the supplied experiment outputs.

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Notation and Abbreviations

OLS	Ordinary Least Squares
SVD	Singular Value Decomposition
QR	Orthogonal–triangular factorisation
VIF	Variance Inflation Factor
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
X	Design matrix
y	Response vector
β	Regression coefficient vector
λ	Ridge regularisation parameter
$\kappa_2(\cdot)$	Matrix 2-norm condition number

Chapter 1

Introduction

1.1 Background and Motivation

Linear regression is a foundational statistical method for modelling the relationship between a response variable and a collection of explanatory variables. In the ordinary least-squares (OLS) framework, the regression coefficients are estimated by minimising the sum of squared residuals [4].

Although the least-squares optimisation problem is mathematically straightforward, the numerical reliability of its solution depends strongly on the properties of the design matrix. When predictor variables exhibit strong linear relationships or near-linear dependence, the estimated regression coefficients can become highly sensitive to small perturbations in the data. This phenomenon is commonly referred to as *multicollinearity*.

From the perspective of numerical linear algebra, the least-squares problem can be solved using different computational approaches, including the normal equations, QR decomposition, and Singular Value Decomposition (SVD). These methods have different numerical properties, particularly when the design matrix is poorly conditioned [1], [6].

Ridge Regression provides a regularised alternative by introducing an L_2 penalty into the least-squares objective. This penalty shrinks coefficient estimates towards zero and can reduce the influence of poorly determined directions in the design matrix [2], [3].

Motivated by these considerations, this study examines the numerical behaviour of different OLS solution methods in the presence of multicollinearity and investigates the effect of Ridge Regression on coefficient stability, matrix conditioning, and predictive performance. The empirical investigation uses the UCI Real Estate Valuation dataset [7], [8].

1.2 Problem Statement

The central problem addressed in this study is to investigate how the numerical properties of a regression design matrix influence coefficient estimation and how regularisation can

modify the stability of the resulting regression system.

The study compares the solutions obtained using normal equations, QR decomposition, and SVD, with particular attention to coefficient estimates, numerical conditioning, and predictive performance.

Furthermore, the study examines whether introducing a ridge penalty reduces coefficient magnitude and improves the conditioning of the regularised system. Five-fold cross-validation is employed to select the regularisation parameter and evaluate the out-of-fold predictive performance of Ridge Regression relative to OLS.

1.3 Objectives

The specific objectives of this study are:

1. Formulate the ordinary least-squares problem and solve it using normal equations, QR decomposition, and Singular Value Decomposition.
2. Compare the coefficient estimates, numerical diagnostics, and predictive performance obtained through the three OLS solution approaches.
3. Investigate multicollinearity using predictor correlations, condition numbers, matrix rank, and singular-value analysis.
4. Implement Ridge Regression and examine the effect of regularisation on coefficient shrinkage and the conditioning of the regression system.
5. Select the regularisation parameter using five-fold cross-validation and compare the validation performance of OLS and Ridge Regression.

1.4 Dataset and Scope

The empirical investigation uses the Real Estate Valuation dataset obtained from the UCI Machine Learning Repository. The dataset contains 414 observations and seven original fields, comprising six explanatory variables and one response variable [7], [8].

The original explanatory variables describe the transaction date, house age, distance to the nearest MRT station, number of nearby convenience stores, latitude, and longitude. The response variable is house price of unit area.

During preprocessing, the transaction-date variable is separated into year and month components for modelling. The regression implementation subsequently constructs an expanded design matrix, including the required encoded variables and an intercept. The initial specification contained 18 columns and was rank deficient. After removing one redundant column, the revised design matrix contained 17 columns and was full rank.

The detailed preprocessing procedure and construction of the regression design matrix are

presented in Chapter 3.

The scope of this study is restricted to the numerical and statistical investigation of OLS and Ridge Regression. Exploratory data analysis is included where it contributes to understanding predictor relationships, multicollinearity, and the numerical properties of the regression design.

1.5 Organisation

The remainder of this report is organised as follows.

Chapter 2 presents the mathematical and theoretical foundations of linear regression, OLS estimation, matrix factorisation methods, multicollinearity, and Ridge Regression.

Chapter 3 describes the dataset, preprocessing procedures, design matrix construction, and computational methodology adopted in the study.

Chapter 4 presents the experimental results, including the comparison of OLS solution methods, multicollinearity diagnostics, Ridge Regression experiments, and cross-validation results.

Chapter 5 discusses the experimental findings, their numerical and statistical implications, and the limitations of the study.

Chapter 6 concludes the study by summarising the principal findings and identifying possible directions for future investigation.

Chapter 2

Theoretical Background

2.1 Linear Regression

Linear regression is a statistical modelling technique used to describe the relationship between a dependent variable and one or more independent variables. It assumes that the response can be represented as a linear combination of the predictors and an error term.

For a dataset containing n observations and p predictors, the multiple linear regression model can be expressed as

$$y = X\beta + \varepsilon, \tag{2.1}$$

where

- $y \in \mathbb{R}^n$ is the response vector,
- $X \in \mathbb{R}^{n \times p}$ is the design matrix,
- $\beta \in \mathbb{R}^p$ is the vector of unknown regression coefficients,
- $\varepsilon \in \mathbb{R}^n$ is the error vector.

The objective of linear regression is to estimate the coefficient vector β such that the predicted response values are close to the observed values.

The predicted response is given by

$$\hat{y} = X\hat{\beta}. \tag{2.2}$$

2.2 Ordinary Least Squares Estimation

Ordinary Least Squares (OLS) estimates the regression coefficients by minimising the sum of squared residuals. The residual vector is defined as

$$r = y - X\beta. \tag{2.3}$$

The OLS objective function is therefore

$$J(\beta) = \|y - X\beta\|_2^2. \tag{2.4}$$

Expanding the objective function gives

$$J(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X\beta. \tag{2.5}$$

Differentiating with respect to β and setting the gradient equal to zero,

$$\frac{\partial J(\beta)}{\partial \beta} = -2X^T y + 2X^T X\beta = 0. \tag{2.6}$$

This leads to the normal equations:

$$X^T X\hat{\beta} = X^T y. \tag{2.7}$$

If $X^T X$ is nonsingular, the OLS estimator is

$$\boxed{\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y} \tag{2.8}$$

The normal equations provide a direct mathematical solution to the least-squares problem. However, explicitly forming and inverting $X^T X$ can be numerically problematic when the design matrix is ill-conditioned. This motivates the use of alternative numerical methods such as QR decomposition and Singular Value Decomposition (SVD) [1], [6].

2.3 Numerical Methods for Solving OLS

2.3.1 Normal Equations

The normal equations provide a direct approach to computing the least-squares solution:

$$X^T X\hat{\beta} = X^T y. \tag{2.9}$$

When $X^T X$ is invertible, the solution is obtained using Equation 2.8.

Although this method is mathematically straightforward, its numerical reliability depends on the conditioning of the design matrix. In particular, the condition number of the Gram matrix satisfies

$$\kappa_2(X^T X) = \kappa_2(X)^2, \tag{2.10}$$

for a full-column-rank matrix X under the spectral norm.

Consequently, forming the Gram matrix can amplify the numerical effects of ill-conditioning. This is particularly relevant when predictors exhibit strong linear relationships.

2.3.2 QR Decomposition

QR decomposition is a numerical technique that factorises a matrix into the product of an orthogonal matrix and an upper triangular matrix.

For a full-column-rank design matrix X, its reduced QR decomposition is

$$X = QR, \tag{2.11}$$

where

- $Q \in \mathbb{R}^{n \times p}$ has orthonormal columns,
- $R \in \mathbb{R}^{p \times p}$ is an upper triangular matrix.

The orthogonality property of Q is expressed as

$$Q^T Q = I. \tag{2.12}$$

Substituting the QR factorisation into the least-squares objective gives

$$\|y - X\beta\|_2^2 = \|y - QR\beta\|_2^2. \tag{2.13}$$

The least-squares solution satisfies

$$R\hat{\beta}_{QR} = Q^T y. \tag{2.14}$$

Thus, the coefficient vector can be obtained by solving the triangular system

$$\boxed{\hat{\beta}_{QR} = R^{-1} Q^T y} \tag{2.15}$$

In practical numerical implementations, the triangular system is solved directly rather than explicitly calculating R^{-1}. QR decomposition generally offers better numerical stability than explicitly forming the normal equations [1], [6].

2.3.3 Singular Value Decomposition

Singular Value Decomposition is a matrix factorisation technique that can be applied to any real matrix, including rectangular and rank-deficient matrices.

The SVD of the design matrix is expressed as

$$X = U\Sigma V^T, \tag{2.16}$$

where

- $U \in \mathbb{R}^{n\times n}$ is an orthogonal matrix,
- $\Sigma \in \mathbb{R}^{n\times p}$ is a diagonal matrix containing the singular values,
- $V \in \mathbb{R}^{p\times p}$ is an orthogonal matrix.

The singular values are non-negative and are conventionally arranged in descending order:

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0, \tag{2.17}$$

where r is the rank of the matrix.

The Moore–Penrose pseudoinverse of X can be constructed from its SVD. For a compact SVD, it is given by

$$X^+ = V\Sigma^+ U^T, \tag{2.18}$$

where Σ^+ is formed by taking the reciprocals of the non-zero singular values and transposing the rectangular diagonal structure.

The least-squares solution is then

$$\boxed{\hat{\beta}_{SVD} = X^+ y} \tag{2.19}$$

SVD provides a useful framework for examining rank deficiency and numerical conditioning. Very small singular values indicate directions in which the design matrix has little numerical information. In practical implementations, a numerical tolerance may be used to determine which singular values are treated as zero.

2.4 Condition Number and Multicollinearity

2.4.1 Condition Number

The condition number measures the sensitivity of a matrix-based problem to small perturbations in its input. For a matrix X, the spectral condition number is defined as

$$\boxed{\kappa_2(X) = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)}} \tag{2.20}$$

where $\sigma_{\max}$ and $\sigma_{\min}$ denote the largest and smallest singular values, respectively, for a full-column-rank matrix.

A large condition number indicates that the matrix is ill-conditioned. In regression, this can make the estimated coefficients sensitive to small changes in the data.

For a full-column-rank matrix, the relationship between the design matrix and its Gram matrix is

$$\kappa_2(X^T X) = \kappa_2(X)^2. \tag{2.21}$$

This relationship explains why solving OLS through the normal equations may be less numerically reliable when the design matrix is poorly conditioned.

2.4.2 Multicollinearity

Multicollinearity occurs when two or more independent variables in a regression model exhibit strong linear relationships.

When predictors are highly correlated, it becomes difficult to distinguish their individual contributions to the response variable. This can lead to unstable coefficient estimates, even when the overall predictive performance of the model remains reasonable.

Some important consequences of multicollinearity include:

- Increased sensitivity of regression coefficients to small changes in the dataset.
- Large standard errors associated with individual coefficient estimates.
- Difficulty in interpreting the individual effects of correlated predictors.
- Numerical instability in the estimation process.

Multicollinearity does not necessarily imply poor predictive performance. However, it can reduce the reliability of coefficient interpretation and make the estimation problem numerically sensitive.

2.4.3 Variance Inflation Factor

The Variance Inflation Factor (VIF) is a diagnostic measure used to quantify the extent to which the variance of a regression coefficient is inflated due to linear relationships among predictors.

For predictor X_j, the VIF is defined as

$$\boxed{VIF_j = \frac{1}{1 - R_j^2}} \tag{2.22}$$

where R_j^2 is the coefficient of determination obtained by regressing predictor X_j on all the remaining predictors.

A VIF value close to one indicates little linear dependence on the other predictors, whereas larger values indicate stronger relationships with the remaining predictors.

VIF is therefore useful as a complementary diagnostic to pairwise correlation and matrix condition numbers.

2.5 Ridge Regression

Ridge Regression is a regularised form of linear regression that introduces a penalty on the magnitude of the regression coefficients. It was introduced to address difficulties associated with highly correlated predictors and unstable coefficient estimates [3].

Unlike OLS, which minimises only the residual sum of squares, Ridge Regression minimises the residual sum of squares together with an L_2 penalty on the coefficients.

The ridge objective function is

$$\boxed{J(\beta) = \|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2} \tag{2.23}$$

where $\lambda \geq 0$ is the regularisation parameter.

A larger value of λ imposes a stronger penalty on coefficient magnitudes, while $\lambda = 0$ reduces the objective to the ordinary least-squares problem.

2.5.1 Ridge Estimator

Differentiating the ridge objective function with respect to β gives

$$\frac{\partial J(\beta)}{\partial \beta} = -2X^T y + 2X^T X\beta + 2\lambda\beta. \tag{2.24}$$

Setting the gradient equal to zero results in

$$X^T X\hat{\beta}_{Ridge} + \lambda\hat{\beta}_{Ridge} = X^T y. \tag{2.25}$$

Rearranging,

$$(X^T X + \lambda I)\hat{\beta}_{Ridge} = X^T y. \tag{2.26}$$

Therefore, the ridge estimator is

$$\boxed{\hat{\beta}_{Ridge} = (X^T X + \lambda I)^{-1} X^T y} \tag{2.27}$$

For $\lambda > 0$, the addition of λI shifts the eigenvalues of the Gram matrix away from zero, making the regularised system better conditioned.

2.5.2 Treatment of the Intercept

In practical regression models, the intercept is generally not penalised. If the first column of the design matrix corresponds to the intercept, the ridge penalty can be represented using the matrix

$$P = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}. \tag{2.28}$$

The corresponding ridge objective becomes

$$J(\beta) = \|y - X\beta\|_2^2 + \lambda \beta^T P \beta. \tag{2.29}$$

The resulting estimator is

$$\boxed{\hat{\beta}_{Ridge} = (X^T X + \lambda P)^{-1} X^T y} \tag{2.30}$$

This formulation ensures that the intercept remains unpenalised while the remaining coefficients are regularised.

2.5.3 Effect of Regularisation on Singular Values

The effect of ridge regularisation can also be understood through the SVD of the design matrix.

Let

$$X = U \Sigma V^T. \tag{2.31}$$

For a centred predictor matrix with all predictor coefficients penalised, the ridge estimator can be expressed as

$$\hat{\beta}_{Ridge} = V(\Sigma^T \Sigma + \lambda I)^{-1} \Sigma^T U^T y. \tag{2.32}$$

Each singular direction is therefore scaled by a factor of the form

$$\frac{\sigma_i}{\sigma_i^2 + \lambda}. \tag{2.33}$$

When a singular value is small, regularisation reduces the influence of the corresponding direction. This helps control coefficient magnitudes and reduces sensitivity to directions associated with weak information in the design matrix.

As λ increases, the coefficient estimates are increasingly shrunk towards zero. However, this introduces bias into the estimates. Ridge Regression therefore involves a trade-off between coefficient stability and model fit [2].

2.6 Selection of the Regularisation Parameter

The performance of Ridge Regression depends on the choice of the regularisation parameter λ. A very small value may provide insufficient regularisation, while an excessively large value may shrink the coefficients too strongly and lead to underfitting.

Cross-validation provides a systematic approach to selecting this parameter.

2.6.1 K-Fold Cross-Validation

In K-fold cross-validation, the available dataset is divided into K approximately equal subsets, known as folds.

The procedure is as follows:

1. Divide the dataset into K folds.
2. Select one fold as the validation set and use the remaining folds for training.
3. Fit the model using the training data.
4. Evaluate the model on the validation fold.
5. Repeat the procedure until each fold has been used once as the validation set.
6. Calculate the average validation error across all folds.

For a given regularisation parameter λ, the mean cross-validation error is

$$CV(\lambda) = \frac{1}{K} \sum_{k=1}^{K} MSE_k(\lambda), \tag{2.34}$$

where $MSE_k(\lambda)$ represents the mean squared error obtained on the k-th validation fold.

The selected regularisation parameter is

$$\boxed{\lambda^* = \underset{\lambda \in \Lambda}{\operatorname{argmin}}\ CV(\lambda)} \tag{2.35}$$

where Λ denotes the set of candidate regularisation parameters.

The use of cross-validation helps evaluate the predictive performance of different regularisation strengths on data not used to fit the corresponding fold-specific model. Preprocessing steps such as feature scaling must also be fitted using only the training portion of each fold to avoid data leakage.

2.7 Performance Evaluation Metrics

The regression models in this study are evaluated using four commonly used performance metrics: Mean Squared Error, Root Mean Squared Error, Mean Absolute Error, and the coefficient of determination.

2.7.1 Mean Squared Error

Mean Squared Error (MSE) measures the average squared difference between the actual and predicted response values.

$$\boxed{MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2.36}$$

A smaller MSE indicates that the predictions are closer to the observed values, with larger errors receiving greater weight due to squaring.

2.7.2 Root Mean Squared Error

Root Mean Squared Error (RMSE) is the square root of MSE:

$$\boxed{RMSE = \sqrt{MSE}} \tag{2.37}$$

RMSE is expressed in the same units as the response variable and provides a measure of the typical magnitude of prediction error, while remaining sensitive to larger errors.

2.7.3 Mean Absolute Error

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values.

$$\boxed{MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|} \tag{2.38}$$

Unlike MSE, MAE does not square the residuals and is therefore less sensitive to large individual errors.

2.7.4 Coefficient of Determination

The coefficient of determination, denoted by R^2, measures the proportion of variation in the response variable explained by the regression model.

It is defined as

$$\boxed{R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{2.39}$$

where $\bar{y}$ is the mean of the observed response values.

An R^2 value closer to one generally indicates a better fit to the observed data. However, the metric should be interpreted alongside error measures and the evaluation procedure used.

2.8 Summary

This chapter presented the theoretical foundations required for the experimental investigation of OLS and Ridge Regression. The normal equations, QR decomposition, and SVD provide alternative approaches to solving the least-squares problem, while condition numbers and VIF help investigate numerical sensitivity and multicollinearity.

Ridge Regression introduces an L_2 penalty to control coefficient magnitudes and improve the conditioning of the regularised system. The selection of its regularisation parameter can be performed using cross-validation, while regression performance can be assessed through MSE, RMSE, MAE, and R^2.

These concepts provide the mathematical basis for the methodology and experimental comparisons presented in the subsequent chapters.

Chapter 3

Dataset and Experimental Methodology

3.1 Dataset Description

The empirical investigation uses the Real Estate Valuation dataset obtained from the UCI Machine Learning Repository. The dataset contains 414 observations from real-estate transactions in the Sindian District of New Taipei City. It consists of seven original variables, comprising six explanatory variables and one response variable [7], [8].

The response variable is the house price of unit area. The explanatory variables describe transaction date, house age, distance to the nearest MRT station, number of nearby convenience stores, latitude, and longitude.

The original variables are summarised in Table 3.1.

Table 3.1: Variables in the original Real Estate Valuation dataset

Symbol	Role	Description
X_1	Predictor	Transaction date (decimal-year representation)
X_2	Predictor	House age (years)
X_3	Predictor	Distance to the nearest MRT station
X_4	Predictor	Number of nearby convenience stores
X_5	Predictor	Geographic latitude
X_6	Predictor	Geographic longitude
Y	Response	House price of unit area

3.2 Preprocessing and Design Matrix Construction

The saved data-processing notebooks transform the decimal transaction-date variable into separate year and month components for modelling.

The regression implementation subsequently expands the date-related components and incorporates an intercept into the design matrix.

The initial regression design matrix contained 18 columns but had a rank of 17. This indicated the presence of one linearly redundant column. To obtain a full-column-rank specification, the redundant column was removed, resulting in a revised design matrix containing 17 columns with rank 17.

This distinction is important because a rank-deficient design matrix does not possess a unique ordinary least-squares coefficient solution through the standard inverse-based normal-equation formulation. Removing the redundant column produces a full-column-rank specification for the subsequent OLS comparison.

Two related but distinct predictor matrices are considered in the experimental analysis:

1. **EDA predictor matrix:** The six numerical predictors are used for preliminary correlation, VIF, and condition-number analysis.
2. **Regression design matrix:** The expanded 17-column design matrix, including the intercept, is used in the OLS implementation and numerical comparison.

The condition numbers calculated from these two matrices are therefore not directly interchangeable. Their values depend on the dimensions, scaling, encoding, and intercept treatment of the matrices being examined.

3.3 OLS Implementations

The ordinary least-squares problem is solved using three computational approaches: normal equations, QR decomposition, and Singular Value Decomposition.

All three approaches are applied to the revised 17-column full-rank regression design matrix.

3.3.1 Normal Equations

The normal-equation approach computes the coefficient estimates by solving

$$(\mathbf{X}^\top \mathbf{X})\widehat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}.$$

The implementation forms the Gram matrix $\mathbf{X}^\top \mathbf{X}$ and solves the resulting linear system.

The numerical properties of this approach are examined through the condition number of the design matrix and its corresponding Gram matrix.

3.3.2 QR Decomposition

The QR implementation factorises the design matrix as

$$\mathbf{X} = \mathbf{QR},$$

where $\mathbf{Q}$ has orthonormal columns and $\mathbf{R}$ is upper triangular.

The coefficient estimates are obtained by solving

$$\mathbf{R}\widehat{\boldsymbol{\beta}} = \mathbf{Q}^\top \mathbf{y}.$$

This approach avoids explicitly forming the Gram matrix and provides an alternative numerical solution to the least-squares problem.

3.3.3 Singular Value Decomposition

The SVD implementation decomposes the design matrix as

$$\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top.$$

The least-squares solution is constructed using the singular values and corresponding orthogonal factors:

$$\widehat{\boldsymbol{\beta}} = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^\top \mathbf{y}.$$

The singular values are also examined to assess the rank and numerical conditioning of the regression design matrix.

3.3.4 Comparison Criteria

The three OLS implementations are compared using the following quantities:

- Estimated regression coefficients.
- Mean Squared Error (MSE).
- Root Mean Squared Error (RMSE).
- Mean Absolute Error (MAE).
- Coefficient of determination (R^2).
- Matrix rank and condition number.
- Residual norms and decomposition diagnostics.

These measures provide a basis for comparing the numerical solutions and their predictive

performance.

3.4 Ridge Regression

Ridge Regression is implemented by introducing an L_2 regularisation penalty into the least-squares objective.

The saved implementation standardises the predictor variables and centres the response variable before applying the regularisation procedure.

The intercept is handled separately and is excluded from the penalty. Consequently, regularisation is applied only to the predictor coefficients.

For the centred and standardised predictor matrix $\mathbf{X}_s$, the ridge estimator is expressed as

$$\widehat{\boldsymbol{\beta}}_{\text{ridge}} = (\mathbf{X}_s^\top \mathbf{X}_s + \lambda \mathbf{I})^{-1} \mathbf{X}_s^\top \mathbf{y}_c,$$

where $\mathbf{y}_c$ denotes the centred response vector and λ is the regularisation parameter.

The intercept is subsequently recovered separately from the fitted predictor coefficients and the centring information.

The regularisation parameter controls the amount of coefficient shrinkage. A range of candidate values is evaluated to investigate its effect on coefficient magnitudes and the conditioning of the regularised system.

Three graphical diagnostics are used:

- **Ridge trace:** Displays the coefficient estimates against different values of the regularisation parameter.
- **Coefficient norm:** Illustrates the reduction in the overall coefficient magnitude as regularisation increases.
- **Condition-number trace:** Examines how the condition number of the regularised system changes with the regularisation parameter.

3.5 Cross-Validation Protocol

Five-fold cross-validation is used to select the regularisation parameter.

The saved cross-validation implementation uses five folds with random state 42. Candidate values of λ are evaluated over a logarithmically spaced parameter grid.

For each fold, the model is fitted using the training partition, and its predictive performance is evaluated on the corresponding held-out validation partition.

Preprocessing operations, including predictor standardisation, are fitted using only the training partition within each fold. The estimated transformation is then applied to the

validation partition. This prevents information leakage from the validation observations into the model-fitting process.

For a candidate regularisation parameter λ, the mean validation error is

$$\text{CV-MSE}(\lambda) = \frac{1}{5}\sum_{k=1}^{5} \text{MSE}_k(\lambda),$$

where $\text{MSE}_k(\lambda)$ denotes the validation mean squared error for fold k.

The selected regularisation parameter is the candidate value that minimises the mean validation error:

$$\lambda^\star = \arg\min_{\lambda \in \Lambda} \text{CV-MSE}(\lambda),$$

where Λ represents the evaluated set of candidate regularisation values.

The OLS baseline is evaluated using the same fold assignments to provide a consistent comparison with Ridge Regression.

3.6 Implementation Environment

The computational experiments are implemented in Python using NumPy and pandas for numerical computation and data manipulation. The generated visualisations are saved as PNG files, while numerical results and diagnostic outputs are exported as CSV files.

The implementation, supporting notebooks, and experimental resources are maintained in the project's public GitHub repository [5]:

`https://github.com/Roaljh/Ridge`

The saved Jupyter notebooks document the data-processing workflow, OLS implementations, Ridge Regression experiments, and cross-validation procedure.

The corresponding experimental outputs are organised separately into data-processing and implementation results, as described in Appendix A.

Chapter 4

Results and Experimental Analysis

4.1 Preliminary Multicollinearity Diagnostics

The preliminary exploratory analysis examines the six numerical predictors to investigate their relationships and assess the presence of multicollinearity.

The condition number of the raw predictor matrix is approximately 237426.0354. After standardisation, the condition number decreases to approximately 4.2097.

The corresponding Gram matrix condition numbers are approximately 5.6371×10^{10} for the raw predictor matrix and 17.7214 for the standardised predictor matrix.

The substantial reduction demonstrates that the numerical conditioning of the raw matrix is strongly influenced by differences in the scales and magnitudes of its predictor variables.

The variance inflation factor (VIF) values range from approximately 1.0055 to 4.2938. The largest VIF is associated with the distance to the nearest MRT station.

Furthermore, the strongest pairwise predictor correlation is observed between distance to the nearest MRT station and longitude, with a Pearson correlation coefficient of $r = -0.8063$.

These diagnostics provide complementary information about predictor relationships. In particular, the pairwise correlation indicates a notable association between MRT distance and longitude, while the VIF values quantify the extent to which each predictor can be explained by the remaining predictors.

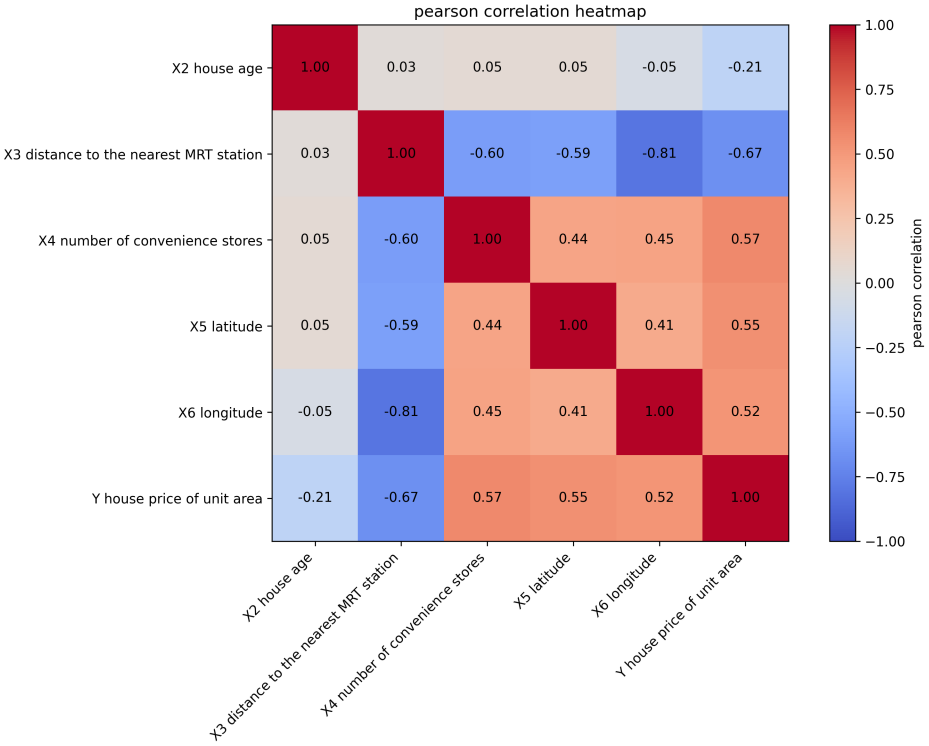


Figure 4.1: Pearson correlation heatmap for the real-estate variables.

4.2 OLS: Normal Equations, QR and SVD

The revised regression design matrix contains 414 observations and 17 columns, with a numerical rank of 17.

The condition number of the revised design matrix is approximately

$$\kappa_2(\mathbf{X}) \approx 2.3840 \times 10^{7}.$$

The corresponding Gram matrix condition number is approximately

$$\kappa_2(\mathbf{X}^\top \mathbf{X}) \approx 5.6835 \times 10^{14}.$$

The smallest singular value of the revised design matrix is approximately

$$\sigma_{\min}(\mathbf{X}) \approx 1.4206 \times 10^{-3}.$$

These diagnostics describe the expanded 17-column regression design matrix and must be distinguished from the six-predictor EDA diagnostics presented in the preceding section.

The three OLS approaches are evaluated using the full dataset. Their corresponding performance metrics are reported in Table 4.1.

Table 4.1: Full-data OLS performance comparison using three solution approaches.

Method	MSE	RMSE	MAE	R^2
Normal equations (revised)	74.9730	8.6587	6.0134	0.5941
QR decomposition	74.9730	8.6587	6.0134	0.5941
SVD	74.9730	8.6587	6.0134	0.5941

The three methods produce effectively identical performance metrics to the reported precision. This agreement is expected because all three approaches solve the same least-squares optimisation problem for the revised full-rank design matrix.

However, their computational procedures differ. The normal-equation approach operates through the Gram matrix, whereas QR decomposition and SVD work directly with the design matrix through orthogonal factorisations.

The QR and SVD implementations also exhibit small decomposition and orthogonality errors. The normal-equation implementation reports a residual norm of approximately 1.1426×10^{-7}. The coefficient and prediction comparisons with the corresponding NumPy reference routines are also small in the saved diagnostics.

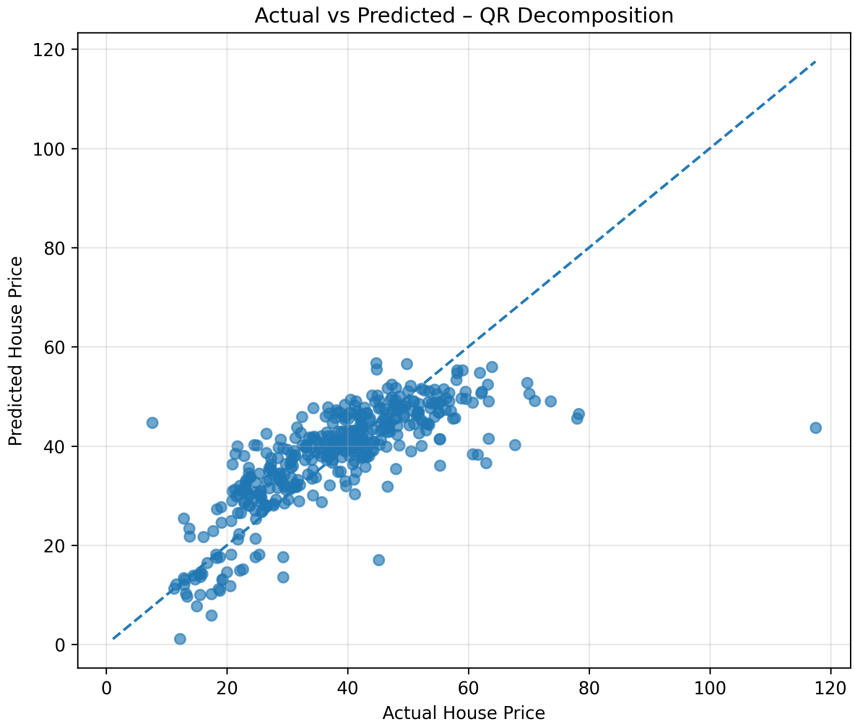


Figure 4.2: Actual versus predicted response obtained using QR decomposition.

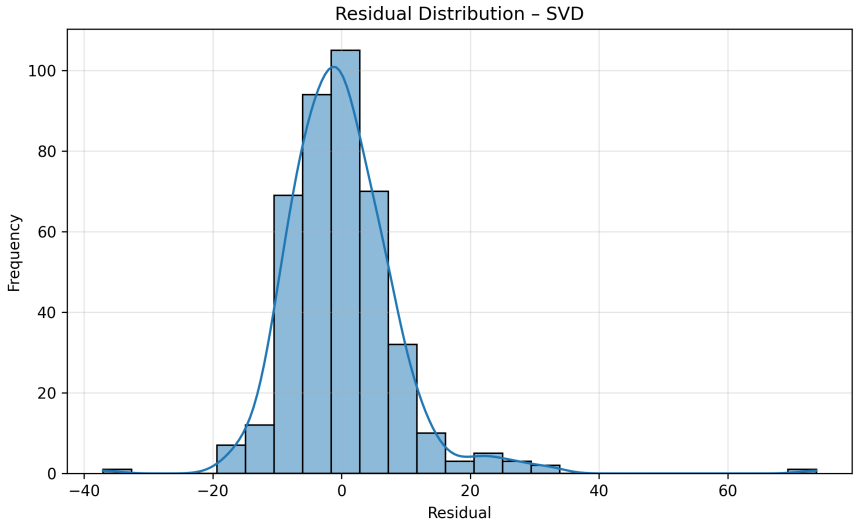


Figure 4.3: Residual distribution obtained using the SVD least-squares solution.

4.3 Ridge Regression and Coefficient Shrinkage

Ridge Regression introduces an L_2 regularisation penalty to control the magnitude of the regression coefficients.

The ridge trace illustrates the evolution of individual coefficient estimates as the regularisation parameter λ increases.

The coefficient paths demonstrate the shrinkage effect introduced by the regularisation penalty. As the penalty increases, the overall magnitude of the coefficient vector decreases.

The coefficient norm provides a complementary quantitative view of this behaviour. Its reduction with increasing λ indicates that the ridge estimator progressively shrinks the predictor coefficients.

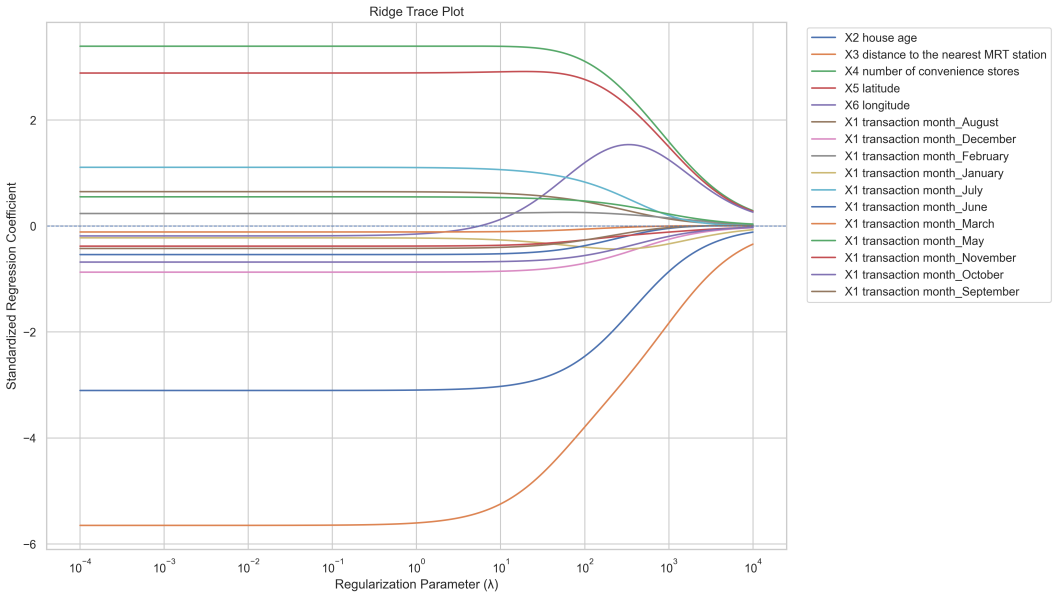


Figure 4.4: Ridge trace showing the evolution of regression coefficients over the regularisation parameter.

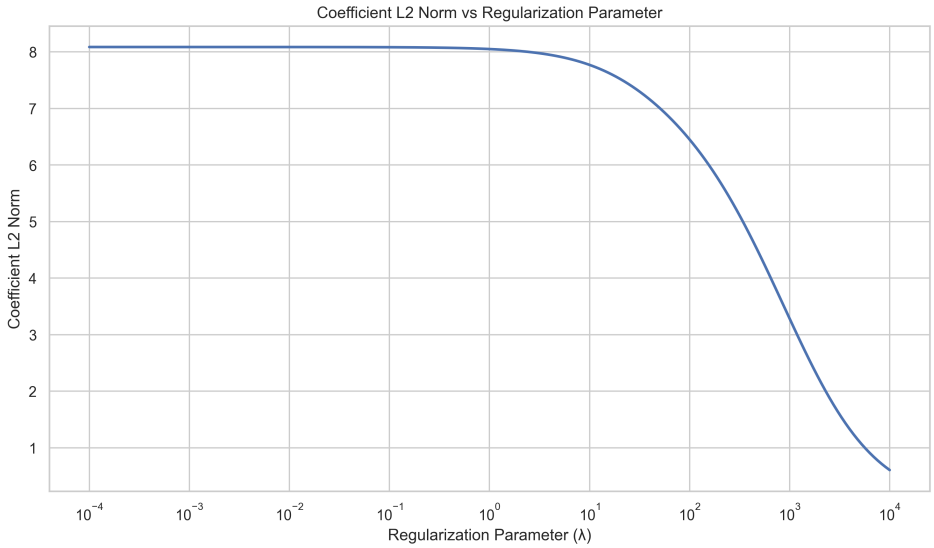


Figure 4.5: Coefficient L_2 norm as a function of the ridge regularisation parameter.

The conditioning of the regularised system is also examined across the evaluated penalty values.

The ridge penalty modifies the eigenvalues of the regularised normal-equation system. Consequently, the condition number changes as λ increases.

The resulting condition-number trace is presented in Figure 4.6.

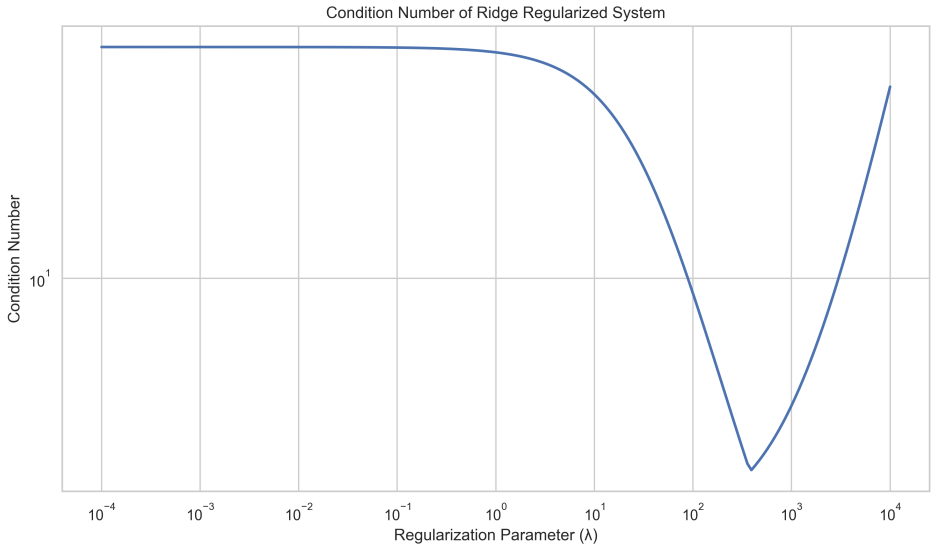


Figure 4.6: Condition number of the ridge-regularised system across the evaluated penalty grid.

4.4 Five-Fold Cross-Validation

Five-fold cross-validation is used to select the regularisation parameter. The experiment uses random state 42 and evaluates candidate values of λ over a logarithmically spaced parameter grid.

The selected regularisation parameter is

$$\boxed{\lambda^\star = 35.297073}.$$

At this value, the mean cross-validated MSE is 86.4162, with a standard deviation of 35.1264. The corresponding mean RMSE, MAE, and coefficient of determination are 9.1674, 6.4478, and 0.5361, respectively.

For comparison, the OLS baseline evaluated on the same fold assignments produces a mean MSE of 87.0110, mean RMSE of 9.2031, mean MAE of 6.4470, and mean R^2 of 0.5323.

The comparison is summarised in Table 4.2.

Table 4.2: Five-fold cross-validation comparison using mean metrics across the validation folds.

Model	MSE	RMSE	MAE	R^2
OLS	87.0110	9.2031	6.4470	0.5323
Ridge ($\lambda^\star = 35.2971$)	86.4162	9.1674	6.4478	0.5361

The mean validation MSE of Ridge Regression is approximately 0.68% lower than that of

OLS in this experiment.

Although Ridge Regression achieves a lower mean validation MSE and RMSE, the mean MAE remains almost unchanged. The improvement in the validation coefficient of determination is also relatively small.

The observed difference should therefore be interpreted as a modest empirical improvement rather than conclusive evidence of a substantial predictive advantage.

The variation in validation performance across folds further indicates that the results should be interpreted with appropriate caution.

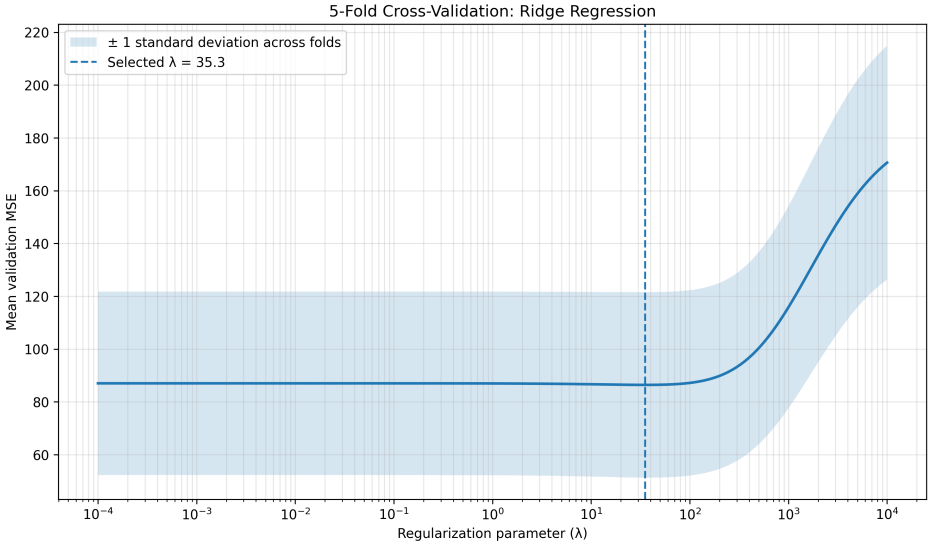


Figure 4.7: Mean five-fold cross-validation MSE over the evaluated ridge penalty grid.

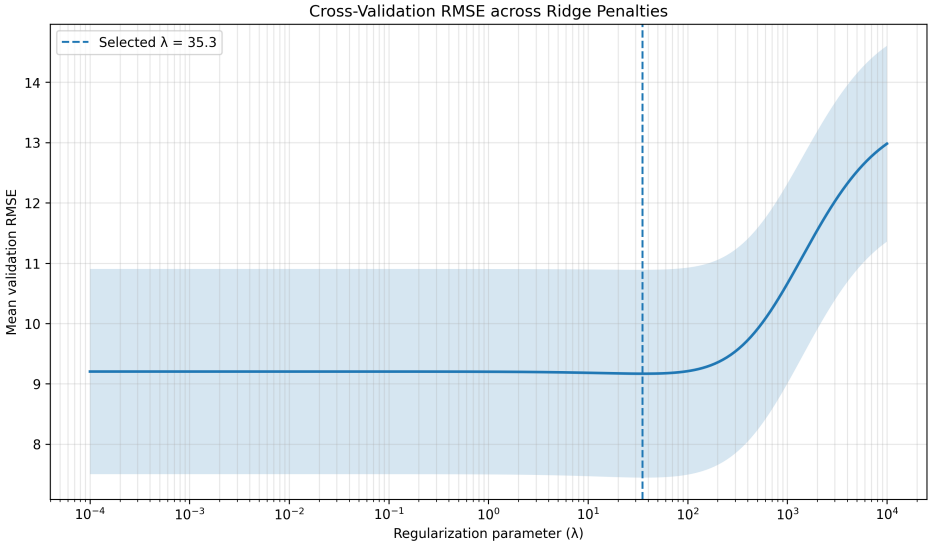


Figure 4.8: Mean five-fold cross-validation RMSE over the evaluated ridge penalty grid.

The predictions obtained using the selected cross-validated ridge model are illustrated in Figure 4.9.

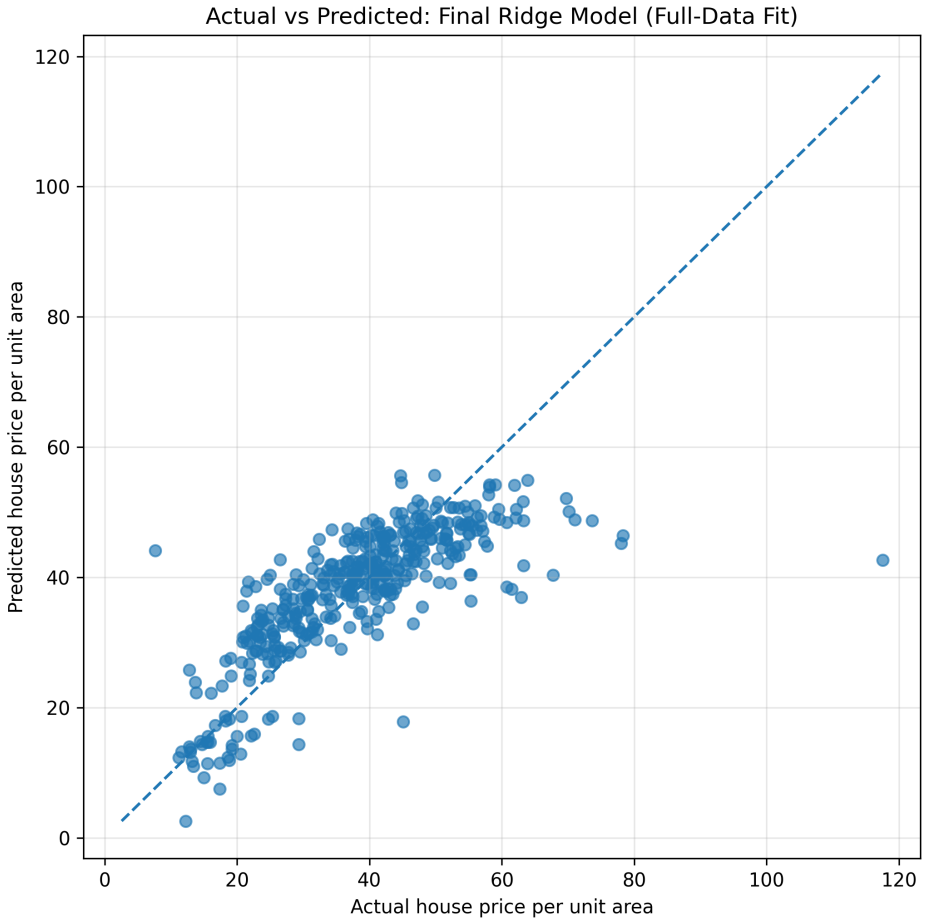


Figure 4.9: Actual versus predicted response for the selected cross-validated Ridge Regression model.

4.5 In-Sample OLS–Ridge Comparison

In addition to the cross-validation experiment, a separate full-data comparison between OLS and Ridge Regression is available in the saved experimental results.

The OLS model reports an MSE of 74.9730, whereas the ridge model with the saved baseline penalty $\lambda = 1$ reports an MSE of 74.9737.

The corresponding performance metrics are presented in Table 4.3.

Table 4.3: Full-data in-sample performance comparison between OLS and Ridge Regression.

Model	MSE	RMSE	MAE	R^2
OLS	74.9730	8.6587	6.0134	0.5941
Ridge ($\lambda = 1$)	74.9737	8.6587	6.0143	0.5941

The two models exhibit nearly identical in-sample performance. The slight increase in the ridge MSE is consistent with the introduction of a regularisation penalty, which constrains

coefficient magnitudes.

This in-sample comparison must be distinguished from the five-fold cross-validation experiment presented in the preceding section. The latter evaluates out-of-fold predictive performance, whereas the present comparison describes the fit obtained using the full dataset.

The two experiments therefore address different aspects of model performance and should not be interpreted as contradictory results.

Chapter 5

Discussion

5.1 Agreement of the OLS Solvers

The normal-equation, QR decomposition, and Singular Value Decomposition implementations produce effectively identical MSE, RMSE, MAE, and R^2 values to the reported precision.

This agreement is expected because all three methods aim to obtain the same least-squares solution when the design matrix has full column rank. However, their computational procedures and numerical properties differ.

The normal-equation approach operates through the Gram matrix $\mathbf{X}^\top\mathbf{X}$, whereas QR decomposition and SVD work directly with the design matrix using orthogonal factorisations.

For a full-column-rank matrix, the condition number of the Gram matrix is the square of the condition number of the original design matrix in the 2-norm. Consequently, forming the normal equations can amplify numerical conditioning difficulties [1], [6].

The agreement in the reported prediction metrics therefore demonstrates consistency among the three implementations for the considered experiment. It does not imply that the three computational approaches have identical numerical stability characteristics.

5.2 Interpreting the Conditioning Diagnostics

The conditioning diagnostics in this study refer to two different matrices: the six-predictor matrix used in exploratory data analysis and the expanded regression design matrix used in the OLS experiments.

For the six numerical predictors, the condition number decreases from approximately 237426.0354 to 4.2097 after standardisation. This substantial reduction indicates that the scaling and numerical magnitudes of the original predictors strongly influence the

conditioning of the raw matrix.

The corresponding maximum VIF is approximately 4.2938, while the strongest pairwise correlation is observed between distance to the nearest MRT station and longitude.

These diagnostics provide evidence of predictor relationships, but they should not be interpreted independently as proof of severe multicollinearity across the entire predictor set.

The expanded regression design matrix presents a different numerical situation. Its condition number is approximately 2.3840×10^7, while the corresponding Gram matrix condition number is approximately 5.6835×10^{14}.

The initial design specification contained 18 columns but had rank 17. The categorical representations of year and month were examined during preprocessing. Retaining both representations resulted in a rank-deficient design. The categorical year representation was therefore removed, while the month representation was retained. The resulting 17-column design matrix was full rank.

This illustrates the importance of examining the rank and conditioning of the actual regression design matrix rather than relying exclusively on pairwise correlations or diagnostics computed from a different predictor representation.

Furthermore, condition numbers depend on the scaling, encoding, and intercept treatment of the matrix being analysed. Consequently, the EDA and regression design diagnostics must be interpreted separately.

5.3 Effect of Ridge Regularisation

Ridge Regression introduces an L_2 penalty that reduces the magnitude of the predictor coefficients.

The ridge trace and coefficient-norm results illustrate this shrinkage behaviour as the regularisation parameter increases.

From a numerical perspective, the regularisation term modifies the eigenvalues of the penalised normal-equation system. This can reduce the influence of directions associated with small singular values and improve the conditioning of the regularised system.

However, the effect of regularisation involves a trade-off between bias and variance. While coefficient shrinkage can reduce sensitivity to poorly determined directions, it also introduces bias into the estimated coefficients [2], [3].

The results demonstrate that Ridge Regression provides a mechanism for controlling coefficient magnitude and modifying the numerical properties of the regression system.

Its effectiveness should nevertheless be assessed using both numerical diagnostics and predictive performance, rather than coefficient shrinkage alone.

5.4 Cross-Validation Interpretation

The five-fold cross-validation experiment selects a regularisation parameter of approximately

$$\lambda^\star = 35.297073.$$

At this value, Ridge Regression achieves a mean validation MSE of 86.4162, compared with 87.0110 for the OLS baseline evaluated on the same fold assignments.

The relative reduction in mean validation MSE is approximately 0.68%.

The mean validation RMSE also decreases slightly, from 9.2031 for OLS to 9.1674 for Ridge Regression. However, the mean validation MAE remains nearly unchanged, with values of 6.4470 and 6.4478, respectively.

The mean coefficient of determination increases from 0.5323 to 0.5361.

These results indicate a small improvement in aggregate validation performance for Ridge Regression under the selected cross-validation configuration.

Nevertheless, the improvement is modest, and the standard deviation of the ridge validation MSE across the five folds is approximately 35.1264. Therefore, the observed difference should not be interpreted as conclusive evidence of a substantial or consistently reproducible predictive advantage.

The results instead suggest that regularisation provides a modest benefit for the particular dataset, preprocessing procedure, and cross-validation configuration considered in this study.

5.5 Limitations

The findings of this study should be interpreted in the context of the following limitations:

1. **Dataset scope:** The experiments are conducted using a single real-estate valuation dataset. Consequently, the findings cannot automatically be generalised to other datasets or regression problems.
2. **Feature representation:** The numerical properties of the regression design depend on the preprocessing and feature-encoding choices. In particular, the removal of the categorical year representation was necessary to obtain the revised full-rank design matrix.
3. **Cross-validation configuration:** The model-selection experiment uses five-fold cross-validation with random state 42 and a specified candidate regularisation grid. Different fold assignments or candidate grids may produce different selected values of λ and validation metrics.

4. **Validation uncertainty:** The observed improvement in mean validation MSE is relatively small. Repeated cross-validation or an independent held-out test set would provide additional evidence about the stability of this improvement.
5. **Computational reproducibility:** Although the source code, notebooks, and experimental outputs are made available through the project repository, an immutable environment lockfile containing exact software versions is not included in the current experimental record. Therefore, exact package-version reproducibility is not asserted.

The study is intended as a numerical and experimental investigation of OLS and Ridge Regression rather than a causal analysis of real-estate prices. Its conclusions are therefore restricted to the observed numerical behaviour and predictive results under the adopted modelling procedure.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

This project investigated the numerical behaviour of Ordinary Least Squares (OLS) and Ridge Regression in the presence of multicollinearity using the UCI Real Estate Valuation dataset. The study focused on the implementation and comparison of three OLS solution methods—normal equations, QR decomposition, and Singular Value Decomposition (SVD)—along with an examination of matrix conditioning and the effect of ridge regularisation.

During the construction of the regression design matrix, the initial representation of both year and month as categorical variables resulted in a rank-deficient matrix. This issue was addressed by removing the categorical year representation while retaining the month representation. The resulting design matrix contained 17 columns and had full column rank.

The three OLS solution methods produced effectively identical results on the full dataset, with an MSE of 74.9730, an RMSE of 8.6587, an MAE of 6.0134, and an R^2 value of 0.5941. This agreement demonstrates that the different numerical approaches yielded consistent solutions for the revised design matrix.

The conditioning analysis highlighted the numerical sensitivity associated with the regression problem. In particular, the condition number of the Gram matrix was substantially larger than that of the corresponding design matrix. This illustrates how forming the normal equations can amplify conditioning-related difficulties. The analysis of pairwise correlations and Variance Inflation Factors (VIFs) also provided additional evidence of relationships among the predictors.

Ridge Regression was subsequently implemented to investigate the effect of regularisation on coefficient magnitudes and numerical conditioning. The regularisation parameter was selected using five-fold cross-validation with a random state of 42, resulting in an optimal value of approximately $\lambda^* = 35.2971$. In the cross-validation comparison, Ridge Regression

achieved a mean validation MSE of 86.4162, compared with 87.0110 for OLS. Although this represents a modest improvement in mean validation MSE, the difference is not large enough to establish a substantial predictive advantage.

Overall, the experiments demonstrate the importance of numerical methods and matrix conditioning in linear regression. They also illustrate the role of ridge regularisation in controlling coefficient magnitudes and improving numerical stability, while introducing a bias–variance trade-off that must be considered when evaluating predictive performance.

6.2 Future Scope

The present study provides several directions for further investigation:

1. **Repeated and nested cross-validation:** Repeated cross-validation could be used to examine the consistency of the selected regularisation parameter and the observed validation performance across different data partitions. Nested cross-validation could provide a more rigorous evaluation of the parameter-selection procedure.
2. **Independent test-set evaluation:** An independent held-out test set could be introduced to assess the generalisation performance of OLS and Ridge Regression on observations not used during model selection.
3. **Alternative feature representations:** The sensitivity of the regression results to different feature-encoding strategies could be investigated, particularly for date-related variables and categorical predictors.
4. **Comparison with other regularisation methods:** Ridge Regression could be compared with LASSO and Elastic Net to study differences in coefficient shrinkage, feature selection, and predictive performance.
5. **Coefficient stability and perturbation analysis:** Controlled perturbations to the input data could be introduced to examine how multicollinearity affects the stability of estimated regression coefficients under OLS and Ridge Regression.
6. **Broader numerical investigation:** Further experiments could examine the influence of matrix conditioning, singular values, and regularisation strength on the numerical behaviour of regression algorithms across different datasets.

These extensions would provide a broader understanding of the numerical and statistical properties of regularised regression methods and help determine the robustness of the observations made in the present study.

Appendix A

Reproducibility and Supplementary Material

A.1 Source Code and Implementation

The computational implementation and experimental workflow developed for this study are publicly available in the project's GitHub repository:

`https://github.com/Roaljh/Ridge`

The repository contains the implementation associated with the data-processing workflow, Ordinary Least Squares (OLS) estimation, QR decomposition, Singular Value Decomposition (SVD), and Ridge Regression experiments. It also provides the supporting computational resources required to understand and reproduce the numerical experiments presented in this report [5].

The implementation primarily uses Python and its numerical computing ecosystem, including NumPy and pandas.

A.2 Dataset Source

The experiments in this study use the Real Estate Valuation dataset obtained from the UCI Machine Learning Repository.

The dataset contains 414 observations and six explanatory variables, with house price of unit area as the response variable. The original dataset and its description are available through the following source:

`https://doi.org/10.24432/C5J30W`

The dataset is cited as Yeh (2018) in the bibliography [8].

A.3 Organisation of Experimental Results

The experimental outputs are organised into two principal categories to distinguish data exploration and preprocessing from the implementation of the regression methods.

A.3.1 Data Processing Results

The `results/data_processing/` directory contains the outputs associated with the preliminary analysis and preprocessing of the dataset.

It is organised as follows:

- `figures/`: visualisations of feature distributions, correlation patterns, and other exploratory data analysis results.
- `tables/`: CSV files containing descriptive statistics, correlation matrices, variance inflation factors, and condition-number diagnostics.

A.3.2 Implementation Results

The `results/implementation/` directory contains the numerical outputs generated during the implementation and comparison of the regression methods.

It is organised as follows:

- `figures/`: plots illustrating OLS comparisons, residual behaviour, ridge coefficient paths, ridge traces, and cross-validation results.
- `tables/`: CSV files containing estimated coefficients, regression metrics, numerical diagnostics, and cross-validation summaries.

A.4 Computational Workflow

The overall computational procedure followed in this study can be summarised in the following stages:

1. **Data preparation:** Loading the Real Estate Valuation dataset, examining the variables, and preparing the design matrix for regression analysis.
2. **Exploratory analysis:** Investigating feature distributions, pairwise correlations, variance inflation factors, and condition numbers to assess potential multicollinearity.
3. **OLS estimation:** Computing the regression coefficients using the normal equations, QR decomposition, and Singular Value Decomposition.
4. **Numerical comparison:** Comparing the coefficient estimates, prediction errors, and numerical stability of the three OLS approaches.
5. **Ridge Regression:** Applying L_2 regularisation and examining the effect of different

regularisation parameters on the coefficient estimates.

6. **Model selection:** Selecting the regularisation parameter λ using five-fold cross-validation.
7. **Evaluation:** Comparing the resulting models using appropriate regression metrics and visual diagnostics.

A.5 Reproducibility

The source code, notebooks, and supporting experimental outputs provide the computational basis for the results reported in this study.

For reproducibility, the following components should be considered together:

- The publicly available GitHub repository containing the implementation.
- The original UCI Real Estate Valuation dataset.
- The saved Jupyter notebooks documenting the computational workflow.
- The exported CSV files containing numerical results and diagnostic information.
- The figures generated during data processing and model implementation.

The numerical results presented in the main chapters correspond to the experimental outputs generated through this workflow. The GitHub repository provides access to the implementation, while the cited UCI source identifies the original dataset.

A.6 Final Project Resources

The principal external resources used in this study are summarised below:

- **Dataset:** UCI Real Estate Valuation Dataset, `https://doi.org/10.24432/C5J30W`.
- **Source code and implementation:** GitHub repository, `https://github.com/Roaljh/Ridge`.

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